

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Operating System

Semester: Fall

Year : 2020
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is Operating System? "Operating system acts as extended machine as well as resource manager", explain this statement. Clarify with the suitable example. 8
b) Differentiate between process and thread. Explain the field in process control block (PCB). 7
2. a) What is semaphore? Explain how you solve producer-consumer problem using semaphore. 7
b) What is IPC? Describe its implementation using shared memory and message passing. 8
3. a) Five Processes and 3 resource types A, B, C and D (Below is the snapshot of the state as: 8

Process	Max	Allocation	Available
	A, B, C, D	A, B, C, D	A, B, C, D
P0	6 0 1 2	4 0 0 1	3 2 1 1
P1	2 7 5 0	1 1 0 0	
P2	2 3 5 6	1 2 5 4	
P3	1 6 5 3	0 6 3 3	
P4	1 6 5 6	0 2 1 2	

- Is this a safe state? If yes, what is safe sequence?
- b) What are different types of threads? Explain context switching with 7

respect to kernel mode and user-mode.

4. a) Given the following set of information, What is the average waiting time and average turn-around time using SJF(Preemptive), FCFS, RR (Quantum = 3) and HRRN.

Process	Arrival Time	Service Time (Burst Time)
A	0	8
B	2	6
C	4	9
D	7	5
E	9	4

- b) Differentiate between logical address and physical address. Explain contiguous and non-contiguous memory allocation approach with their advantages and disadvantages.
5. a) Consider the following page reference strings: 2,3,4,5,3,2,6,7,3,2,3,4,8,7,4,3,2,3,4,7. How many page faults would occur for each of the following page replacement algorithms assuming 3 pages a frame? In each case calculate fault ratio.

i) Optimal Page Replacement

ii) LRU page replacement

iii) FIFO page replacement

- b) Suppose a disk drive has cylinders numbered from 0 through 3999. The drive is currently serving a request at cylinder 299. The queue of pending request in FIFO order is given by 916,1509,82,1011,1774,130,507,250,2681,56. Calculate total distance (in cylinders) in FCFS, SSF and SCAN. Which one is best?

- a) Describe Access Control Matrix and Access Control List using an appropriate example. How it achieves a level of security in files?
- b) Define distributed operating system. Explain Remote Procedure Call (RPC) with the help of an appropriate figure.

Write short notes on: (Any two)

- a) Operating system structure
- b) Clock synchronization in DS
- c) LINUX operating system

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Attempt all the questions.

- a) What is an operating System? Discuss the main advantage for an operating system designer for using virtual machine architecture. Give the main advantage for user. 8
- b) Write advantages of threads over processes. How multiprocessor system can be effectively utilized with threads? Explain with example. 7
- a) What is test and set instruction? Explain producer-consumer problem and algorithm for resolving it using monitor. 8
- b) Consider the deadlock situation that could occur in the dining philosophers' problem when the philosophers obtain the chopstick one at a time. Discuss how the four necessary conditions for deadlock indeed hold in this setting. Discuss how deadlock could be avoided by eliminating any one of the four conditions. 7
- a) Write advantages of threads over processes. Explain the advantage of multithreading. 8
- b) Given the following information, draw the GANTT charts for processor scheduling for HRRN, Preemptive Shortest Job First and RR (Quantum=2). Also, find the average waiting time, average turnaround time and average response time for all the cases. 7

Process	Arrival Time	Burst Time
P1	0.0	7
P2	3.0	4
P3	5.0	2
P4	6.0	4

4. a) What is thrashing? Consider the following page reference strings: 2,3,3,4,5,6,5,7,1,2,5,8,6,4,1. How many page faults would occur for each of the following page replacement algorithms assuming 3 pages frames?

- i. LRU page replacement
- ii. FIFO page replacement
- iii. Optimal page replacement

b) Define swapping. Explain contiguous and non-contiguous memory allocation scheme with their advantages and disadvantages.

5. a) Suppose a disk drive has 2000 cylinders, numbered 0 to 1999. The drive is currently serving a request at cylinder 134 and the previous request was at cylinder 124. The queue of pending requests in FIFO order is 86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130. Starting from the current head position what is the total distance that the disk arm moves to satisfy all the pending requests for each of the following disk scheduling algorithms?

- i) FCFS ii) SSTF iii) SCAN iv) CSCAN

b) What is file system implementation? Explain link list and i-node file system implementations.

a) What is distributed operating system? Explain advantage of distributed system over independent PC.

b) Explain file and disk management in LINUX.

Write short notes on: (Any two)

- a) Context Switching
- b) Internal vs External Fragmentation
- c) Multilevel feedback queues

POKHARA UNIVERSITY

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Semester: Spring

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Attempt all the questions.

1. a) Explain briefly about Types of OS along with its evolution in brief. 8
b) What is a process? Explain various states and transitions between the states of process. 7
2. a) What is a deadlock? Explain with example dead lock avoidance for multiple resources using banker's algorithm. 8
b) What is "Race Condition"? Illustrate its effect on the execution of a system with an example? 7
3. a) Give the structure of a kernel? How kernel mode and user mode are different? 7
b) Five batch jobs A through E, arrive at a computer center at almost same time. They have estimated running times of 10, 6, 2, 4 and 8 minutes. Their priorities are 3,5,2,4 and 1 respectively with 5 being the highest priority. For each of the scheduling algorithms determine the mean process turn around time. Ignore process switching overhead 8
 - i) Round robin scheduling (quantum time = 2 minutes)
 - ii) Priority scheduling
 - iii) Shortest job first
4. a) Consider the following page reference strings: 1, 3, 5, 3, 7, 1, 5, 3, 1, 2, 3, 7, 6, 3, 4, 1, 8. How many page faults would occur for each of the following page replacement algorithms assuming 4 pages a frames? 7
 - i. LRU page replacement
 - ii. FIFO page replacement
 - iii. Optimal page replacement

- b) What is swapping? Given Memory partition of 100k, 500k, 200k, 300k and 600k in order. How would first fit algorithm place processes of 212k, 417k, 112k and 426k in order?
5. a) The disk track reques are 123,250,298,120,13,300 and 220.
Assume that the last request is at track 150 and the head is moving towards track 0. Find out the local seek time for each of the link scheduling algorithm below. 7
- i) SSF
 - ii) C-SCAN
 - iii) FIFO
- b) Explain different file system implementation methods.
6. a) List the goals of distributed operating system and explain in detail. How does RPC communicate in distributed environment?
- b) Describe the use and importance of autoexec.bat and config.sys files in MS-DOS.
7. Write short notes on: (Any two)
- a) Parallel OS.
 - b) Ostrich Algorithm
 - c) DMA

POKHARA UNIVERSITY

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Semester: Fall

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Attempt all the questions.

1. a) Define processes. Explain different operating system structures. 8
b) What is process control block? Describe the fields in a process control block (PCB) with diagram. 7
2. a) Explain Producer Consumer problem in process synchronization and give solutions to it using semaphores. 7
b) Describe Dining Philosophers Problem and show how deadlock occurs in it. Provide the solution to deadlock in Dining Philosophers problem. 8
3. a) List out the conditions for deadlock. Explain deadlock modelling using resource allocation graphs for multiple type Resources. 8
b) What is context switching? Explain it with an appropriate diagram. 7
4. a) Consider the following set of processes that arrive at time 0, with the length of the CPU burst given in milliseconds. Construct the Gantt chart and calculate average waiting time. 8

Process	Burst Time	Priority
P1	32	3
P2	7	6
P3	6	2
P4	21	1
P5	6	4
P6	9	5

- i. HRRN
 - ii. RR (Quantum = 5)
 - iii. Priority Scheduling
 - iv. Shortest Job First (Preemptive)
- b) List pros and cons of distributed system over centralized system. Explain remote procedure call (RPC)
5. a) What is fragmentation? What are its types? How can the problems of fragmentation be solved?
- b) Consider the following page reference strings 7,0,1,2,0,3,0,4,2,4,0,3,2. How many page faults would occur for each of the following page replacement algorithm assuming 3 pages a frame? In each case calculate fault ratio.
- i. Second Chance Page replacement algorithm
 - ii. NRU
 - iii. Optimal Page replacement algorithm.
6. a) Suppose a disk drive has 200 tracks, numbered 0 to 199. The current position of the R/W head is at track 60 and the previous request was at track 45. The sequence of pending requests is 43, 72, 150, 48, 85, 170, 190, 130. Starting from current position what is the total number of track movements (distance) for the following disk scheduling algorithms.
- i. SSTF
 - ii. SCAN
 - iii. C-LOOK
- b) List the various file operations. Explain the I-nodes method of File allocation.
7. Write short notes on: (Any two)
- a) File Systems and Disk management in LINUX
 - b) Security in Windows 2000.
 - c) RAID.

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Attempt all the questions.

1. a) Define operating system. Explain the fundamental differences between monolithic and layered structure of operating system. 8
b) For what purpose semaphores are used? Give solution to producer-consumer problem using semaphores. 7
2. a) Define PCB. How do processes transition between different states and what triggers these transitions? Illustrate with neat diagram. 8
b) What are different types of kernels? Explain each in detail. 7
3. a) Consider the following set of processes with the length of the CPU burst given in milliseconds. Construct the Gantt chart and calculate average waiting time and turnaround time. Also explain which one is the best algorithm. 8

Process	Arrival Time	Burst Time	Priority
P1	0	9	2
P2	4	4	1
P3	10	3	3
P4	0	6	4

- i) HRRN
 - ii) RR (Quantum = 1ms)
 - iii) Priority Scheduling (1 higher priority)
 - iv) Shortest Job First (pre-emptive)
- b) Examine the role of virtual memory. How does the Operating system map virtual address to physical address? Explain with example. 7

4. a) Consider the following page reference strings: 9, 3, 4, 5, 3, 9, 6, 7, 3, 9, 3, 4, 8, 7, 4, 3, 9, 3, 4, 7. How many page faults would occur for each of the following page replacement algorithms assuming 3 pages a frame? In each case calculate fault ratio.
- i) Second Chance page replacement
 - ii) LRU page replacement
 - iii) FIFO page replacement
- b) Define context switching Explain context switching by demonstrating a diagram that show how CPU switches from one process to another.
5. a) Given a disk drive with 400 cylinders. The driver is currently serving a request at cylinder 162 and previous request was at cylinder 128. The queue of pending request in FIFO order is: 90, 150, 386, 94, 187, 48, 278, 202, 188, and 135. Starting from the current head position what is the total distance in cylinders that the disk to satisfy all the pending request for each of the following disk scheduling algorithms?
- v) FCFS
 - vi) SSTF
 - vii) SCAN
 - viii) C-Look
- b) How file naming is done? Discuss different file allocation methods.
6. a) What are advantages of distributed system? Discuss RPC technique in Distributed System.
- b) How process management is done in UNIX or Linux? Explain. Why is Linux considered more secured then other operating system?
7. Write short notes on: (Any two)
- a) DMA.
 - b) Windows 2000.
 - c) The Shell.

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